LUIS **MATIAS**

luisdafmatias@gmail.com | (+351) 925 224 998 | 20/06/1994 | Category B | Slovakia
Portuguese (Native) | English (C1) | lmatias.com | /luisdafmatias | from Portugal

Skills/Tools SolidWorks, Solid Edge, Teamcenter, 3D Printing, Excel, Ansys, Abaqus, Matlab, Python

Interests CAD, Mechanical Design, DFM, DFA, CNC Machines, 3D Printing, Automation, Programming, Product Development, Additive Manufacturing, Manufacturing Engineer

WORK

SKARTEK ○ **Mechanical Design Engineer** Vlčkovce, Slovakia **Aug 2025 – Present**

- Designing conceptual mechanical solutions to secure customer buy-in and formalize service contracts.
- DFM & Process Optimization: Applying Design for Manufacturing principles across CNC machining, welding, laser cutting and 3D Printing to optimize part geometry and ensure structural durability.
- Collision Mitigation & System Adaptation: Re-engineering active machine assemblies to resolve mechanical interference and integrate customer-driven modifications, ensuring zero-collision performance and enhanced operational safety.
- PLM & CAD Integration: Utilizing Solid Edge for precision modeling and Teamcenter for robust version control, BOM management, and cross-departmental collaboration.

Forvia ○ **VIE Development Engineer/Manufacturing Engineer** Trnava, Slovakia **May 2023 – Dec 2024**

- Working in Instrument Panel programs: Citroën C3 and Opel Frontera.
- Define and articulate the Engineering Levels required for production, serving as a key liaison between R&D and the plant team to ensure seamless collaboration.
- Validate mockups for both interim and permanent solutions, ensuring functionality and compliance with design standards.
- Prepare production documents including Visual BOM and 3D models in PLM system.
- Assist manufacturing in defining the sequence of assembly operations during the initial stages of the conveyor production process, ensuring efficiency and accuracy.
- Independently develop 3D printing solutions for mockups, workstation and conveyor modifications, and production support tools, enhancing functionality and efficiency.
- Develop Excel VBA scripts to streamline stock tracking and ensure accurate inventory.

Forvia ○ **VIE Program Engineer** Gorzów Wielkopolski, Poland **Aug 2022 – May 2023**

- Working in Door Panel programs: Jeep Avenger, Fiat 600 and Alfa Romeo Junior.
- Support quality department in capabilities studies.
- Support suppliers in Gage R&R studies to ensure RPS system definition.
- Support in the Meisterbock build.

INEGI ○ **Research Fellow** Porto, Portugal **Dec 2021 – July 2022**

Project: “Dealing with CT Cardiac Imaging Using Python: An Approach for Future Hemodynamic Simulations”

- Computer Vision - Using Python for 3D reconstruction of heart arteries. Region Growing method.
- User interface with PyQt.

Paper: “Dealing with CT Cardiac Imaging Using Python: An Approach for Future Hemodynamic Simulations”
Conference Paper. M2D2022 - 9th International Conference on Mechanics and Materials in Design (pp. 1109-1114)

STUDIES

University of Porto ○ **M.Sc. Mechanical Engineer** Porto, Portugal **July 2022**

Thesis: “Semi-Automatic Method of Stent Development in Coronary Arteries with Stenosis - Hemodynamic Simulations”

- Semi-automatic method to create arteries with stents using SolidWorks and Creo based on CT images.
- Hemodynamics simulations in ANSYS Fluent with stent applied in the arteries.

Paper: “Semi-Automatic Method of Stent Development for Hemodynamic Simulations in Patient Coronary Arteries with Disease”
Theoretical Analyses, Computations and Experiments of Multiscale Materials - Advanced Structured Materials, Springer
DOI: 10.1007/978-3-031-04548 6_16

MY PROJECTS

CNC Milling Machine **Feb 2026 – Now**

- Under development in design phase.
- Designing in SolidWorks.

CNC Diode Laser/Engraver **Jan 2025 – March 2025**

- Finished project and working machine.
- Designed in SolidWorks.
- Pinion and Belt driven system with NEMA 17 stepper motors. V-Slot profile frame. 3 Axis with autofocus.

EXPERIENCES

Elephant Nature Park ○ **Volunteering | Elephant Caretaker** Chiang Mai, Thailand **2017 (1 Week)**

- Taking care of abandoned and abused elephants and dogs.

RMUTR ○ **Internship | Professor Assistant** Phutthamonthon, Thailand **2017 (3 Months)**

- Support students in their school projects.
- Teach Mathematics and Solid Mechanics to the students.

CERTIFICATIONS

SolidWorks CAD Design Professional [CSWP]

May 2025

Solid Edge Professional Level

April 2026